Formalization of Constrained Flow Matching v1

Project Notes

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1 Goal and Setting

We want to learn a velocity field that transports p_0 to p_1, while enforcing intermediate-time moment constraints. In v1, the state is $x \in \mathbb{R}^2$.

Given

- Initial and terminal marginals p_0, p_1.
- A coupling over endpoint pairs (x_0, x_1) (configured as exact discrete OT or random permutation per batch).
- Constraint times $\mathcal{T}_c = \{0.25, 0.5, 0.75\}$.
- Target moments m_k at each $t_k \in \mathcal{T}_c$.

Wanted Learn:

- a path-correction model $g_\theta(t, x_0, x_1)$,
- a velocity model $v_\psi(t, x)$,

such that induced paths satisfy moment constraints and v_ψ matches path velocities.

2 Reference: Standard CFM

Standard conditional flow matching uses a fixed interpolation path x_t (often linear) and target velocity $u_t = \partial_t x_t$, then trains

$$\mathcal{L}_{\mathrm{CFM}}(\psi) = \mathbb{E}_{(x_0,x_1),\,t}\left[\|v_\psi(t, x_t) - u_t\|^2\right].$$

3 Path Family Used in This Project

3.1 Baseline Path

$$\ell_t(x_0, x_1) = (1-t)x_0 + tx_1.$$

3.2 Constrained-Correction Path

We use an endpoint-preserving correction:

$$x_t = \ell_t + a(t)\, g_\theta(t, x_0, x_1), \qquad a(t) = t(1-t).$$

Because $a(0) = a(1) = 0$, endpoints are exactly preserved: $x_{t=0} = x_0$, $x_{t=1} = x_1$.

3.3 Target Velocity for CFM in Constrained Mode

$$u_t = \partial_t x_t = (x_1 - x_0) + (1-2t)\, g_\theta(t, x_0, x_1) + a(t)\, \partial_t g_\theta(t, x_0, x_1).$$

This is exactly what is computed in code using autograd for $\partial_t g_\theta$.

4 Moment Constraints

4.1 Feature Map Used in v1

For a batch $\{x^{(i)}\}_{i=1}^B \subset \mathbb{R}^2$:

$$\hat{\mu} = \frac{1}{B}\sum_{i=1}^{B} x^{(i)}, \qquad \hat{\Sigma} = \frac{1}{B}\sum_{i=1}^{B} \big(x^{(i)} - \hat{\mu}\big)\big(x^{(i)} - \hat{\mu}\big)^\top.$$

Define

$$\phi(\{x^{(i)}\}_{i=1}^B) = \begin{bmatrix} \hat{\mu} \\ \mathrm{vec}(\hat{\Sigma}) \end{bmatrix} \in \mathbb{R}^6.$$

4.2 Residual at Constraint Times

For each $t_k \in \mathcal{T}_c$:

$$c_k(\theta) = \phi\Big(\{x_{t_k}^{(i)}(\theta)\}_{i=1}^B\Big) - m_k.$$

Small $\|c_k\|$ means constraints are well satisfied.

5 How Target Moments m_k are Built in v1

In the benchmark, (p_0, p_1) are Gaussians:

$$p_0 = \mathcal{N}(\mu_0, \Sigma_0), \qquad p_1 = \mathcal{N}(\mu_1, \Sigma_1).$$

Define the Gaussian OT map matrix:

$$A = \Sigma_0^{-1/2}\left(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\right)^{1/2}\Sigma_0^{-1/2}.$$

Define

$$B_t = (1-t)I + tA, \qquad s(t) = 1 + \kappa\, t(1-t).$$

Then target mean/covariance are:

$$\mu_t = (1-t)\mu_0 + t\mu_1, \qquad \Sigma_t = s(t)^2\, B_t \Sigma_0 B_t^\top.$$

Finally:

$$m_k = \begin{bmatrix} \mu_{t_k} \\ \mathrm{vec}(\Sigma_{t_k}) \end{bmatrix}.$$

6 Loss Terms Used in the Current Implementation

6.1 Path Regularizer

$$R(\theta) = \alpha\,\mathbb{E}\big[\|u_t - (x_1 - x_0)\|^2\big] + \beta\,\mathbb{E}\big[\|\partial_t u_t\|^2\big]\,.$$

Interpretation

- First term keeps the learned path velocity near linear transport velocity.
- Second term penalizes rapid temporal variation of velocity (temporal smoothness).

6.2 Augmented Lagrangian for Path Learning (Stage A/C)

$$\mathcal{L}_g(\theta, \lambda) = R(\theta) + \sum_k \lambda_k^\top c_k(\theta) + \frac{\rho}{2} \sum_k \|c_k(\theta)\|^2.$$

Multiplier updates (with clipping in code):

$$\lambda_k \leftarrow \mathrm{clip}(\lambda_k + \rho\, c_k(\theta),\, -\lambda_{\max},\, \lambda_{\max})\,.$$

Algorithm 1: Mini-batch augmented-Lagrangian updates (code-equivalent)

Inputs: constraint times $\mathcal{T}_c$, target features $\{m_k\}_{t_k \in \mathcal{T}_c}$, hyperparameters $(\alpha, \beta, \rho, \eta_{\text{joint}}, \lambda_{\max})$, and step counts (N_A, N_B, N_C).

1. Initialize multipliers: $\lambda_k \leftarrow 0$ for all $t_k \in \mathcal{T}_c$.
2. **Stage A (optimize θ only):** for $s = 1, \ldots, N_A$:
   (a) Sample paired mini-batch $\{(x_0^{(i)}, x_1^{(i)})\}_{i=1}^B$.
   (b) Sample $t^{(i)} \sim \mathrm{Uniform}(0, 1)$ and compute constrained velocity targets $u_{t^{(i)}}(\theta)$.
   (c) Compute regularizer
   $$R(\theta) \leftarrow \alpha\,\frac{1}{B} \sum_i \|u_{t^{(i)}} - (x_1^{(i)} - x_0^{(i)})\|_2^2 + \beta\,\frac{1}{B} \sum_i \|\partial_t u_{t^{(i)}}\|_2^2.$$
   (d) For each $t_k \in \mathcal{T}_c$, compute
   $$x_{t_k}^{(i)} \leftarrow (1 - t_k) x_0^{(i)} + t_k x_1^{(i)} + t_k(1 - t_k)\, g_\theta(t_k, x_0^{(i)}, x_1^{(i)}),$$
   $$\hat{\mu}_k \leftarrow \frac{1}{B} \sum_i x_{t_k}^{(i)}, \qquad \hat{\Sigma}_k \leftarrow \frac{1}{B} \sum_i (x_{t_k}^{(i)} - \hat{\mu}_k)(x_{t_k}^{(i)} - \hat{\mu}_k)^\top,$$
   $$c_k(\theta) \leftarrow \begin{bmatrix} \hat{\mu}_k \\ \mathrm{vec}(\hat{\Sigma}_k) \end{bmatrix} - m_k.$$
   (e) Assemble AL objective
   $$\mathcal{L}_g \leftarrow R(\theta) + \sum_k \lambda_k^\top c_k(\theta) + \frac{\rho}{2} \sum_k \|c_k(\theta)\|_2^2.$$
   (f) Apply Adam step on θ with gradient $\nabla_\theta \mathcal{L}_g$.

(g) Update multipliers for all $t_k \in \mathcal{T}_c$:

$$\lambda_k \leftarrow \mathrm{clip}(\lambda_k + \rho\, c_k(\theta), -\lambda_{\max}, \lambda_{\max})\,.$$

3. **Stage B (freeze θ, optimize ψ):** for $s = 1, \ldots, N_B$, sample a mini-batch and apply Adam step on ψ with $\nabla_\psi \mathcal{L}_{\mathrm{CFM}}$.

4. **Stage C (joint optimize (ψ, θ)):** for $s = 1, \ldots, N_C$:

   (a) Sample a mini-batch and recompute $\mathcal{L}_g(\theta, \lambda)$ exactly as in Stage A.

   (b) Compute $\mathcal{L}_{\mathrm{CFM}}(\psi; \theta)$ and
   $$\mathcal{L}_{\mathrm{joint}} \leftarrow \mathcal{L}_{\mathrm{CFM}} + \eta_{\mathrm{joint}} \mathcal{L}_g.$$

   (c) Apply Adam step on (ψ, θ) with gradient $\nabla_{(\psi,\theta)} \mathcal{L}_{\mathrm{joint}}$.

   (d) Update all multipliers with the same clipped rule:
   $$\lambda_k \leftarrow \mathrm{clip}(\lambda_k + \rho\, c_k(\theta), -\lambda_{\max}, \lambda_{\max})\,.$$

6.3 CFM Loss for Velocity Network

$$\mathcal{L}_{\mathrm{CFM}}(\psi; \theta) = \mathbb{E}_{(x_0, x_1), t} \left[\| v_\psi(t, x_t(\theta)) - u_t(\theta) \|^2 \right].$$

6.4 Joint Finetune Objective (Stage C)

$$\mathcal{L}_{\mathrm{joint}}(\psi, \theta, \lambda) = \mathcal{L}_{\mathrm{CFM}}(\psi; \theta) + \eta_{\mathrm{joint}}\, \mathcal{L}_g(\theta, \lambda).$$

7 Current Optimization Loop (Exactly What Code Does)

Sampling per Step

At each step:

1. Sample endpoint mini-batch from p_0 and p_1.
2. Solve exact discrete OT assignment on this batch to obtain paired $(x_0^{(i)}, x_1^{(i)})$.
3. Sample $t \sim \mathrm{Uniform}(0, 1)$ for CFM/regularizer terms.

Stage A: Pretrain Path Model g_θ

For N_A steps:

1. Compute $\mathcal{L}_g(\theta, \lambda)$.
2. Gradient step on θ (Adam, lr_g).
3. Update $\lambda_k \leftarrow \lambda_k + \rho c_k$ (with clipping).

Stage B: Train Velocity Model v_ψ with Frozen g_θ

For N_B steps:

1. Freeze θ.
2. Compute $\mathcal{L}_{\text{CFM}}(\psi; \theta)$.
3. Gradient step on ψ (Adam, lr_v).

Stage C: Joint Finetune

For N_C steps:

1. Unfreeze both θ, ψ.
2. Compute $\mathcal{L}_{\text{joint}}$.
3. Gradient step on (θ, ψ).
4. Update multipliers λ_k again.

Baseline Mode

In baseline mode there is no g_θ:

$$x_t = \ell_t, \qquad u_t = x_1 - x_0.$$

Only Stage B is optimized.

8 Why This is More Complex than "Path First, then Flow Matching"

Your suggested two-step idea is:

1. Learn paths (e.g., with a smoothness/Lipschitz regularizer),
2. then train flow matching on those paths.

Current implementation adds two extra mechanisms:

- **Augmented Lagrangian** instead of fixed penalties, to push constraints more aggressively.
- **Stage C joint finetune**, to co-adapt path model and velocity model after separate training.

So, yes: the present loop is strictly more complicated than a plain two-stage approach. It can improve constraint handling, but it also introduces additional instability/tuning sensitivity.

9 How Reported Metrics Are Computed

This section defines the fields written in `metrics.json` and `comparison.json`.

Constraint metrics

For each constrained time $t_k \in \mathcal{T}_c$, compute residual vector

$$c_k = \phi\Big(\{x_{t_k}^{(i)}\}_{i=1}^{B}\Big) - m_k.$$

Then:

$$\texttt{constraint_residual_norms}[t_k] = \|c_k\|_2.$$

The summary scalar is:

$$\texttt{constraint_residual_avg} = \frac{1}{|\mathcal{T}_c|} \sum_{t_k \in \mathcal{T}_c} \|c_k\|_2.$$

Validation CFM metric

On an evaluation batch:

1. sample paired (x_0, x_1) via the configured coupling mode,
2. sample $t \sim \text{Uniform}(0, 1)$,
3. build (x_t, u_t) from current mode (baseline or constrained),
4. evaluate

$$\texttt{cfm_val_loss} = \frac{1}{B} \sum_{i=1}^{B} \left\| v_\psi(t_i, x_{t_i}^{(i)}) - u_{t_i}^{(i)} \right\|_2^2.$$

Intermediate-time Wasserstein metric

For each constrained time $t_k \in \mathcal{T}_c$:

1. Sample $x_0^{(i)} \sim p_0$, $i = 1, \ldots, N$.
2. Integrate the learned velocity field from 0 to t_k with Euler to obtain $\hat{x}_{t_k}^{(i)}$.
3. Compute empirical generated moments:

$$\hat{\mu}_{t_k} = \frac{1}{N} \sum_{i=1}^{N} \hat{x}_{t_k}^{(i)}, \qquad \hat{\Sigma}_{t_k} = \frac{1}{N} \sum_{i=1}^{N} (\hat{x}_{t_k}^{(i)} - \hat{\mu}_{t_k})(\hat{x}_{t_k}^{(i)} - \hat{\mu}_{t_k})^\top.$$

4. Let target moments be $(\mu_{t_k}, \Sigma_{t_k})$ from the analytic Gaussian bridge.

The reported per-time metric uses Gaussian W_2 distance:

$$\texttt{intermediate_w2_gaussian}[t_k] = W_2\Big(\mathcal{N}(\hat{\mu}_{t_k}, \hat{\Sigma}_{t_k}), \mathcal{N}(\mu_{t_k}, \Sigma_{t_k})\Big),$$

with closed-form

$$W_2^2 = \|\hat{\mu}_{t_k} - \mu_{t_k}\|_2^2 + \text{Tr}\left(\hat{\Sigma}_{t_k} + \Sigma_{t_k} - 2\left(\Sigma_{t_k}^{1/2} \hat{\Sigma}_{t_k} \Sigma_{t_k}^{1/2}\right)^{1/2}\right).$$

The summary scalar is:

$$\texttt{intermediate_w2_gaussian_avg} = \frac{1}{|\mathcal{T}_c|} \sum_{t_k \in \mathcal{T}_c} \texttt{intermediate_w2_gaussian}[t_k].$$

Sample-based empirical OT variant To support non-Gaussian settings, we also report a sample-based empirical W_2 proxy:

1. Generate trajectory samples $\{\hat{x}_{t_k}^{(i)}\}_{i=1}^N$ by integrating v_ψ from 0 to t_k.
2. Draw target samples $\{y_{t_k}^{(j)}\}_{j=1}^N$ from the reference distribution at t_k.
3. Compute the optimal bipartite matching $\pi^\star$ minimizing squared Euclidean cost:

$$\pi^\star = \arg\min_{\pi \in S_N} \sum_{i=1}^N \left\| \hat{x}_{t_k}^{(i)} - y_{t_k}^{(\pi(i))} \right\|_2^2.$$

The reported per-time empirical metric is:

$$\texttt{intermediate_empirical_w2}[t_k] = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\| \hat{x}_{t_k}^{(i)} - y_{t_k}^{(\pi^\star(i))} \right\|_2^2}.$$

and the average is:

$$\texttt{intermediate_empirical_w2_avg} = \frac{1}{|\mathcal{T}_c|} \sum_{t_k \in \mathcal{T}_c} \texttt{intermediate_empirical_w2}[t_k].$$

In code this uses exact discrete OT matching (Hungarian algorithm), so evaluation sample count is configured separately to control CPU cost.

Path energy proxy

Using the same evaluation batch/time samples as `cfm_val_loss`:

$$\texttt{path_energy_proxy} = \frac{1}{B} \sum_{i=1}^B \|u_{t_i}^{(i)}\|_2^2.$$

This is a diagnostic quantity, not an objective minimized directly at evaluation time.

Transport quality metrics

To measure endpoint transport induced by v_ψ:

1. Sample $x_0^{(i)} \sim p_0$, $i = 1, \ldots, N$.
2. Integrate forward with explicit Euler over n_{steps} steps:

$$x_{n+1}^{(i)} = x_n^{(i)} + \Delta t \, v_\psi(t_n, x_n^{(i)}), \quad \Delta t = \frac{1}{n_{\text{steps}}}, \quad t_n = n\Delta t.$$

3. Let $\hat{x}_1^{(i)} = x_{n_{\text{steps}}}^{(i)}$.
4. Compute empirical endpoint statistics:

$$\hat{\mu}_1 = \frac{1}{N} \sum_{i=1}^N \hat{x}_1^{(i)}, \qquad \hat{\Sigma}_1 = \frac{1}{N} \sum_{i=1}^N (\hat{x}_1^{(i)} - \hat{\mu}_1)(\hat{x}_1^{(i)} - \hat{\mu}_1)^\top.$$

Then:

$$\texttt{transport_mean_error_l2} = \|\hat{\mu}_1 - \mu_1\|_2,$$
$$\texttt{transport_cov_error_fro} = \|\hat{\Sigma}_1 - \Sigma_1\|_F,$$
$$\texttt{transport_score} = \texttt{transport_mean_error_l2} + \texttt{transport_cov_error_fro}.$$

Stage-history diagnostics

During training, the history log also records:

- `avg_residual_norm`: mini-batch average of $\|c_k\|_2$ over constrained times,
- `loss`: stage objective value,
- and stage-specific terms (e.g., regularizer or `cfm_loss` in Stage C).

10 Example Experiment Settings (Current Repository)

This section summarizes the experiment families currently implemented and logged in this repository.

A. Primary trained benchmark: 2D Gaussian → 2D Gaussian

- **Data family**: both endpoints are Gaussians in $\mathbb{R}^2$.
- **Targets**: intermediate moment targets are analytic Gaussian bridge moments at $t \in \{0.25, 0.50, 0.75\}$.
- **Pair coupling options**: exact discrete OT (`data=gaussian_ot`) or random permutation pairing (`data=gaussian_random`).
- **Training modes**: baseline (linear path), constrained (Stage A/B/C), and comparison (run both).
- **Typical profiles**: `train=default`, `train=ab_only` (Stage C disabled), and `train=smoke`.
- **Representative output roots**: `outputs/hparam_sweep/`, `outputs/ab_only_sweep/`, and timestamped comparison folders under `outputs/<date>/...`.

B. Bridge-SDE preview: Gaussian start with bridge-like stochastic dynamics

- **Purpose**: geometry/diagnostic preview prior to training on more complex non-Gaussian trajectories.
- **Initial law**: Gaussian $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$ from bridge-preview parameters.
- **Simulated dynamics** (from `bridge_sde.py`):

$$dX_t = v_x\, dt + \sigma_x\, dW_t^{(x)},$$
$$dY_t = -\kappa_b\, g(X_t)\, Y_t\, dt + \sigma_y(1 - \delta_b\, g(X_t))\, dW_t^{(y)}, \qquad g(x) = \exp\left(-\frac{(x-c)^2}{2w^2}\right).$$

- **Current status**: used in notebook/preview artifacts (snapshots, spread plots, animation), not yet integrated as a full training target in the main Stage A/B/C experiment pipeline.
- **Output root**: `outputs/preview_bridge_sde/<timestamp>/`.

Scope note At present, the optimization formalized in Sections 1–9 corresponds to the Gaussian-to-Gaussian benchmark pipeline. The Bridge-SDE path is currently a validated simulation/visualization track used to prepare future non-Gaussian training settings.

11 Practical Hyperparameters in the Loop

Symbol / Config	Role
α	weight of velocity-deviation regularizer
β	weight of temporal smoothness term
ρ	augmented-Lagrangian quadratic and multiplier update scale
η_{joint}	weight of $\mathcal{L}_g$ inside Stage C objective
lr_g, lr_v	learning rates for path and velocity networks
N_A, N_B, N_C	number of steps in each stage